

Multi-view relational evidential c-medoid clustering with adaptive weighted

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Abstract—Relational data, where objects are defined by similarities or dissimilarities, is omnipresent and essential in real clustering applications. In this context, relational clustering, which aims to identify groups of similar objects based on their mutual relationships, has become a necessity. However, most existing relational clustering methods cannot effectively handle multi-view data sets while representing uncertainty and imprecision when faced with objects in overlapping clusters. To address this gap, we introduce a new relational clustering method, called Multi-View Evidential C-Medoid clustering with adaptive weightings (MECMdd). Our approach is based on the theory of belief functions to characterize the partial knowledge in cluster assignment. It integrates view weight assignments, estimated locally for each cluster and globally in a collaborative learning framework. We have evaluated our proposition via several experiments using different real-world datasets, compared to other related and state-of-the-art methods, in terms of their advantages and overall effectiveness.

Index Terms—Evidential clustering, credal partition, relational multi-view, belief function

I. INTRODUCTION

Clustering is a fundamental technique in data analysis that aims to identify intrinsic structures within a set of objects. These objects, usually characterized by matrices, sequences, time series, images, texts, etc., have three common representations on which clustering can be based: vector data, symbolic data and relational data. The most common case of relational data is when we have (a matrix of) dissimilarity data, noted $\mathcal{R} = [\tau_{ij}]$, where τ_{ij} is the pairwise dissimilarity (often a distance) between objects i and j . Relational clustering allows pattern recognition based on the dissimilarity matrix $\mathcal{R}$. It aims to identify groups of similar objects based on their mutual relationships, as captured by dissimilarity measures. Relational clustering is particularly useful when computing the distance between objects is complex, when the distance measure cannot be expressed in a simple mathematical way, or when clusters of similar objects cannot be efficiently represented by a single prototype, such as a centroid for example.

Numerous relational clustering techniques have been proposed in the literature. Among efficient approaches, the

family of partition-based methods, including the popular k-means algorithm and its relational variant Partitioning Around Medoids (PAM) [1], has been widely used because they are relatively simple to compute, easy to interpret, and they scale efficiently with large data sets. The k-means algorithm, along with its relational versions, identifies clusters using a centroid, respectively a medoid, within a hard partition, where each object is unambiguously assigned to only one cluster. However, in many real-world applications, object data can be ambiguous or uncertain and under such conditions, using hard clustering may result to low clustering accuracy. To address this issue and capture the degree of ambiguity, soft clustering variants have been proposed. These variants allow the expression of uncertainty and/or imprecision in the partition.

Fuzzy c-means (FCM) [2], as the core paradigm of soft clustering, can represent uncertainty by allowing an object to belong to different clusters with various degrees of fuzziness. Several relational variants of the FCM clustering algorithm have been proposed [3]–[5], the most popular is fuzzy c-medoid (FCMdd) [6]. While fuzzy clustering approaches have shown promising results, they have poor robustness against noise and outliers, and they cannot explicitly describe imprecision in the result of cluster assignment. However there is often the need to express both uncertainty and imprecision in clustering. As illustrated in Figure 1, objects between or among different clusters are indistinguishable and can be considered imprecise and/or uncertain. To address such issue, an evidential clustering version of FCM, referred to as evidential c-means (ECM), has been proposed [7]. Based on the belief functions theory [8] and by generating a credal partition, ECM enhances hard and fuzzy partitions [9], while expressing with more precision the magnitude of partial knowledge concerning the class membership of the various objects. ECM offers robust and efficient relational variants based on centroids, such as RECM [10] and EVCLUS [11], as well as variants based on medoids like ECMdd [12], [13], and others based on the median, such as MECM [14].

This work on relational evidential clustering solves the

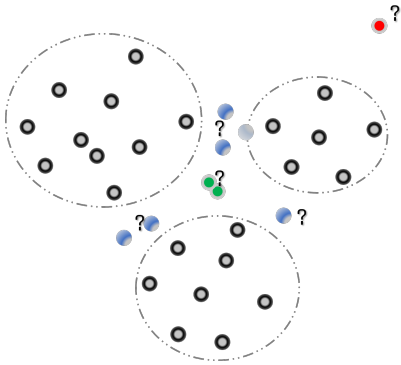


Fig. 1: Illustration of imprecision and uncertainty in clustering.

problems encountered in fuzzy clustering, i.e. it can express in a rich manner the imprecision and uncertainties regarding the class membership of the objects, and allow a direct representation of outliers. Unfortunately, these clustering methods are based on the assumption of single-view data. In several real-world applications, data have multiple representations or sources, which usually have complementary information and can be used to improve the accuracy of the clustering task [15]. Therefore, it is essential to integrate these views to generate more robust and accurate clustering, instead of considering only a single data view. In multi-view case, the traditional approach is to cluster one global matrix that combines the partial similarities in a uniform way. Unfortunately, this approach may not be effective since the influence of the different similarity matrices is generally not equally important in the definition of the category to which similar patterns belong [16]. Therefore, to obtain meaningful clusters across all similarity matrices, we need to learn cluster-dependent relevance weights for each similarity matrix. In evidential clustering context, this problem has recently been addressed for vector data by simultaneously performing clustering and feature discrimination [17], [18]. However, it was ignored for the case of relational data.

In this paper, we propose a new medoid-based multi-view relational clustering algorithm (MECMdd), which is an extension of conventional ECM. In addition to characterizing uncertainty and imprecision in cluster assignment, MECMdd learns a relevance weight for each matrix in each cluster, by optimizing an objective function.

The rest of this paper is organized as follows. Sections II and III present, respectively, previous work on relational multi-view clustering and the general context of evidential clustering. In Section IV, a detailed presentation of the proposed MECMdd clustering algorithm is provided. The performance and the effectiveness of the new algorithms are analyzed using synthetic and real datasets in Section V. Finally, Section VI exposes a conclusion and perspectives about the work.

II. RELATED WORKS ON RELATIONAL MULTI-VIEW CLUSTERING

In this work [19], F. de Carvalho and al. propose MRDCA, an extension of dynamic hard clustering for relational data [20], which uses multiple dissimilarity matrices. However, MRDCA assumes a hard relationship between objects and clusters, which is unrealistic for real data that often contains uncertainty, imprecision, and inconsistency [21]. In [16], Frigui and al. introduced the CARD algorithm based on fuzzy theory, which aggregates multiple dissimilarity matrices and learns a relevance weight for each matrix in each cluster. F. de Carvalho and al. [22], [23] proposed MRFCMdd and its extension MVFCMddV, which builds on the FCMdd algorithm [24]. These methods provide both a partition and prototypes (with MVFCMddV offering vector-based prototypes) for each fuzzy cluster, while simultaneously learning relevance weights both locally and globally.

Recently, a multi-view clustering approaches based on ECM have been proposed [17], [18]. Liu and al. [17] introduced WMVEC, an adaptive weighted multi-view evidential clustering method. However, WMVEC and WMVEC-FP assume access to multi-view vector data. This paper proposes a new adaptive weighted multi-view evidential relational medoid-based clustering (MECMdd) method.

III. BACKGROUND

A. Theory of belief functions

The theory of belief functions [8], [25], also called the Dempster-Shafer theory, has been widely applied in various fields such as classification [26], [27] and clustering [28], [29]. It provides powerful tools for modeling uncertain and imprecise information. Let ω be a variable that takes values in a finite set $\Omega = \{\omega_1, \dots, \omega_c\}$. Partial knowledge of the actual value of ω can be represented by a mass function m , defined from the set 2^Ω to $[0, 1]$ and satisfying: $\sum_{A \subseteq \Omega} m(A) = 1$. The quantity $m(A)$ is then interpreted as the amount of belief allocated to any subset $A \subseteq \Omega$, and A is called a focal element if $m(A) > 0$. When $m(A) = 1$, we have a certain belief. When $m(A) > 0$ and $|A| > 1$, we have an imprecise belief.

For decision making, it is possible to transform a normal mass function into a pignistic probability distribution [30]:

$$BetP(\omega) = \sum_{A \in \mathcal{A}} \frac{m(A)}{|A|} \quad \forall \omega \in \Omega. \quad (1)$$

As an illustration, suppose we have a set of four objects to classify into 2 clusters. Table I presents a credal partition. The class of the first object is certain, while that of the third object is entirely imprecise. The mass function m_2 follows a Bayesian distribution. Regarding the fourth object, it is treated as an outlier since the entire mass is allocated to the empty set.

TABLE I: Example of a credal partition.

A	$m_1(A)$	$m_2(A)$	$m_3(A)$	$m_4(A)$
$\emptyset$	0	0	0	1
$\{\omega_1\}$	1	0.9	0	0
$\{\omega_2\}$	0	0.1	0	0
Ω	0	0	1	0

B. Evidential c-medoids - ECMdd

The ECMdd algorithm is a variant of ECM [12], [13]. In ECMdd, medoids are used as the prototypes to represent the identified clusters, including specific clusters and imprecise subsets. Specific clusters contain data points clearly separated from other cluster prototypes, indicating a clear and distinct classification. Imprecise clusters include data points near multiple prototypes, reflecting classification uncertainty. Let $E = \{e_i \mid i = 1, 2, \dots, n\}$ be a set of n objects, where the only required information is the pairwise dissimilarity matrix $[\tau(e_i, e_j)]$. The objective function is defined as follows:

$$J_{ECMdd}(\mathbf{M}, \mathbf{V}) = \sum_{i=1}^n \sum_{A_j \neq \emptyset} |A_j|^\alpha m_{ij}^\beta \tau_{ij} + \sum_{i=1}^n \delta^2 m_{i\emptyset}^\beta. \quad (2)$$

where m_{ij} is the belief assignment of e_i associated with the nonempty set A_j , $m_{i\emptyset}$ represent the belief assignment of e_i assigned to the empty set, and τ_{ij} is the dissimilarity between e_i and the focal set A_j . Parameters α, β, δ are adjustable and have the same meanings as those in ECM.

IV. EVIDENTIAL MULTIPLE RELATIONAL CLUSTERING USING MEDOIDS

In this section, we introduce a new adaptive weighted relational multi-view evidential clustering (MECMdd) method that simultaneously learns both the consensus credal partition and the significance of each relational view. We consider $v_k^\Omega \in E$ to be the medoid of a specific cluster, where the focal element is a singleton, i.e. $|A_j| = 1$, with $A_j = \{\omega_k\}$ for $k = 1, \dots, c$. When $|A_j| > 1$, representing an imprecise cluster, $v_j^{2^\Omega} \in E$ is defined as the medoid with $A_j \subseteq \{\omega_1, \dots, \omega_c\}$ for $j = 1, \dots, 2^c - 1$. The dissimilarity τ_{ij} is defined by (3) for $|A_j| > 1$, and when $|A_j| = 1$, we have $v_j^{2^\Omega} \triangleq v_k^\Omega$ thus $\tau_{ij} = \tau(x_i, v_k^\Omega)$.

$$\tau_{ij} = \frac{\tau(x_i, v_j^{2^\Omega}) + \gamma \frac{1}{|A_j|} \sum_{\omega_k \in A_j} \tau(x_i, v_k^\Omega)}{1 + \gamma}. \quad (3)$$

The parameter γ quantifies the impact of uncertainty on the dissimilarity between objects and imprecise clusters.

A. MECMdd with relevance weight for each dissimilarity matrix estimated locally

This algorithm is designed to provide a credal partition and a prototype for each credal cluster, while also learning a relevance weight for each dissimilarity matrix. These weights are estimated locally, changing at each iteration of

the algorithm, and vary from one cluster to another.

1) *Formulation:* Let $E = \{e_1, \dots, e_n\}$ be the set of n objects and p dissimilarity matrices defined by $\mathcal{R} = [\tau_l(e_i, e_j)]$ for $l = 1, \dots, p$. The relational evidential c-medoids clustering algorithm with relevance weights for each dissimilarity matrix estimated locally (referred to as MECMdd-RWL) aims to find the partition $\mathbf{M}$, the corresponding prototypes $\mathbf{V}$, and compute the vector of relevance weights (one for each cluster) $\mathbf{\Lambda} = (\lambda_1, \dots, \lambda_k)$, such that $\lambda_k \in \mathbb{R}^p$. It seeks to locally optimize an adequacy criterion (objective function) measuring the fit between the partition and their prototypes. The objective function of MECMdd-RWL can be represented as follows:

$$J_{MECMdd-RWL}(\mathbf{M}, \mathbf{V}, \mathbf{\Lambda}) = \sum_{i=1}^n \sum_{A_j \neq \emptyset} |A_j|^\alpha m_{ij}^\beta \sum_{l=1}^p (\lambda_{jl})^s \tau_{ijl} + \sum_{l=1}^p \delta_l^2 \sum_{i=1}^n m_{i\emptyset}^\beta. \quad (4)$$

Where $\sum_{l=1}^p (\lambda_{jl})^s \tau_{ijl}$ represents the global relation between an object $e_i \in E$ and the subset prototype $v_j^{2^\Omega} \in E$, parameterized by $s > 1$. λ_{jl} is the view weight of p -th view in the subset A_j . $\tau_{ijl} = \tau_l(e_i, A_j) = \tau_l(e_i, v_j^{2^\Omega})$ is the local dissimilarity between $e_i \in A_j$ and the subset prototype $v_j^{2^\Omega} \in E$ in dissimilarity matrix $\tau_l (l = 1, \dots, p)$. The parameter α , and β are the same as in ECMdd. The parameter δ works as a threshold, assumed at a fixed distance from any individuals, at which objects can be considered outliers, and controls the rate of individuals considered outliers. Motivated by the noise clustering (NC) approach, presented as robust to atypical individuals [31], and by the adaptive weighted multi-view evidential clustering proposal [32], the threshold δ based on several dissimilarity matrices is calculated as follows $\sum_{l=1}^p \delta_l^2 \sum_{i=1}^n m_{i\emptyset}^\beta$. The objective function $J_{MRFCMdd-RWL}$ is optimized under the following constraints:

$$\sum_{A_j \subseteq \Omega, A_j \neq \emptyset} m_{ij} + m_{i\emptyset} = 1, \quad \forall i = 1, \dots, n; \quad (5)$$

$$m_{ij} \geq 0 \quad \forall i = 1, \dots, n, \quad \forall A_j \subseteq \Omega; \quad (6)$$

$$\lambda_{jl} \in [0, 1], \quad \sum_{l=1}^p \lambda_{jl} = 1, \quad \forall A_j \subseteq \Omega. \quad (7)$$

2) *Optimization:* This algorithm like ECMdd, starts with an initial partition and alternates three steps until convergence. We provide an efficient iterative method to minimize (4), constrained with (5), (6), and (7). More specifically, we employ the Lagrange optimization method to alternately update one variable while leaving others fixed.

Updating $\mathbf{M}$, with $\mathbf{V}$ and $\mathbf{\Lambda}$ fixed:

Theorem IV.1. *The credal partition, denoted as $\mathbf{M}$, which minimizes the clustering objective function $J_{MECMdd-RWL}$ w.r.t constraints (5) and (6) is computed using the following expression*

$$m_{ij} = \frac{(|A_j|^\alpha \sum_{l=1}^p (\lambda_{jl})^s \tau_{ijl})^{-\frac{1}{\beta-1}}}{\sum_{A_k \neq \emptyset} (|A_k|^\alpha \sum_{l=1}^p (\lambda_{kl})^s \tau_{ikl})^{-\frac{1}{\beta-1}} + (\sum_{l=1}^p \delta_l^2)^{-\frac{1}{\beta-1}}} \quad (8)$$

$$m_{i\emptyset} = 1 - \sum_{A_j \neq \emptyset} m_{ij} \quad (9)$$

Proof. To prove the Theorem IV.1, we employ Lagrange multipliers, denoted as ξ , to tackle the constrained minimization problem of the J-MECMdd-RWL objective function concerning $\mathbf{M}$. The J-MECMdd-RWL function can be reformulated to

$$\mathcal{L}(\mathbf{M}, \boldsymbol{\xi}) = J_{MECMdd-RWL}(\mathbf{M}, \mathbf{V}, \boldsymbol{\Lambda}) - \sum_{i=1}^n \xi_i \left(\sum_{\substack{A_j \subseteq \Omega, \\ A_j \neq \emptyset}} m_{ij} + m_{i\emptyset} - 1 \right) \quad (10)$$

By differentiating the Lagrangian with respect to the m_{ij} , $m_{i\emptyset}$ and ξ_i , we obtain the derivatives (11), (12) and (13). Solving these equations we obtain the solutions (8) and (9).

$$\frac{\partial L(\mathbf{M}, \xi_i)}{\partial m_{ij}} = \beta m_{ij}^{\beta-1} |A_j|^\alpha \sum_{l=1}^p (\lambda_{jl})^s \tau_{ijl} - \xi_i = 0 \quad (11)$$

$$\frac{\partial L(\mathbf{M}, \xi_i)}{\partial m_{i\emptyset}} = \beta m_{i\emptyset}^{\beta-1} \sum_{l=1}^p \delta_l^2 - \xi_i = 0 \quad (12)$$

$$\frac{\partial L(\mathbf{M}, \xi_i)}{\partial \xi_i} = \sum_{A_j \subseteq \Omega, A_j \neq \emptyset} m_{ij} + m_{i\emptyset} - 1 = 0 \quad (13)$$

Updating $\boldsymbol{\Lambda}$, with $\mathbf{M}$ and $\mathbf{V}$ fixed:

Theorem IV.2. The vectors of relevance weights $\boldsymbol{\lambda}_k = (\lambda_{k1}, \dots, \lambda_{kp})$, which minimizes the clustering objective function (4), with $s > 1$, and constraints (7), according to:

$$\lambda_{jl} = \frac{\left(\sum_{i=1}^n s |A_j|^\alpha m_{ij}^\beta \tau_{ijl} \right)^{-\frac{1}{s-1}}}{\sum_{h=1}^p \left(\sum_{i=1}^n s |A_j|^\alpha m_{ij}^\beta \tau_{ijh} \right)^{-\frac{1}{s-1}}} \quad (14)$$

Proof. In order to prove Theorem IV.2, we employ the Lagrange multiplier, denoted as ξ_j , to solve the constrained minimization problem with respect to $\boldsymbol{\Lambda}$. The objective function (4) is written as follows :

$$L(\boldsymbol{\Lambda}, \boldsymbol{\xi}) = J_{MECMdd-RWL}(\mathbf{M}, \mathbf{V}, \boldsymbol{\Lambda}) - \sum_{A_j} \xi_j \left(\sum_{l=1}^p \lambda_{jl} - 1 \right) \quad (15)$$

By differentiating the Lagrangian with respect to $\boldsymbol{\Lambda}$ and ξ_j variables setting the derivatives to zero, we obtain :

$$\frac{\partial L(\boldsymbol{\Lambda}, \xi_j)}{\partial \lambda_{jl}} = \sum_{i=1}^n s |A_j|^\alpha m_{ij}^\beta (\lambda_{jl})^{s-1} \tau_{ijl} - \xi_j = 0 \quad (16)$$

$$\frac{\partial L(\boldsymbol{\Lambda}, \xi_j)}{\partial \xi_j} = \sum_{l=1}^p \lambda_{jl} - 1 = 0 \quad (17)$$

Using (16) and (17), we can isolate the expression of $(\xi_j)^{\frac{1}{s-1}}$ and obtain the updating (14).

Updating $\mathbf{V}$ while fixing $\mathbf{M}$ and $\boldsymbol{\Lambda}$:

Theorem IV.3. The prototype of the singleton cluster $e_p = v_k^\Omega (k = 1, \dots, c)$ of subset $A_j = \{\omega_k\}$ with $e_l \in E$, which minimizes the clustering objective function J-MECMdd-RWL, is first computed according to (18). Next, the prototype of the imprecise cluster $e_l = v_j^{\Omega_j}$ of subset $A_j \subseteq \{\omega_1, \dots, \omega_k\}$ with $e_l \in E$ is determined according to equations (19) and (20).

$$l = \operatorname{argmin}_{1 \leq h \leq n} \sum_{i=1}^n m_{ij}^\beta \sum_{l=1}^p (\lambda_{jl})^s \tau_l(e_i, e_h) \quad (18)$$

$$l = \operatorname{argmin}_{1 \leq h \leq n} \sigma_{ij}^2 + \eta \frac{1}{|A_j|} \sum_{\omega_k \in A_j} \sum_{l=1}^p \tau_l(e_i, v_k^\Omega), \quad \text{and} \quad (19)$$

$$\sigma_{ij}^2 = \frac{1}{|A_j|} \sum_{\omega_k \in A_j} \left[\sum_{l=1}^p \tau_l(e_i, v_k^\Omega) - \frac{1}{|A_j|} \sum_{\omega_k \in A_j} \sum_{l=1}^p \tau_l(e_i, v_k^\Omega) \right]^2 \quad (20)$$

The parameter η , which can typically be set to 1, helps to identify outliers from potential medoids when determining prototypes of meta-classes and has minimal effect on the final partitioning results.

Proof. The proof of Theorem IV.3 is straightforward according to [12].

3) *Computational algorithm and complexity:* The MECMdd-RWL algorithm converges after a fixed number of iteration steps. Indeed, updating the credal partition $\mathbf{M}$ and the local relevance weight vector $\boldsymbol{\Lambda}$ via the Lagrangian does not increase the $J_{MECMdd-RWL}$ function. Also $J_{MECMdd-RWL}$ will not increase through the medoid-searching scheme for prototypes of specific clusters. When the prototypes of singleton clusters are fixed, the medoids of imprecise subsets determined by (19) are expected to be positioned close to the centroid of the prototypes of the singletons clusters [12]. Additionally, since prototypes are assumed to belong to the original object dataset E , there exists a finite number of distinct prototype vectors $\mathbf{V}$ and, consequently, a finite number of corresponding credal partitions $\mathbf{M}$. Therefore, we can infer that the MECMdd-RWL algorithm converges within a finite number of steps. For ease of implementation, we provide a summary of the proposed method in Algorithm 1.

In [13], the algorithmic complexity of ECMdd was analyzed. It was shown that one iteration of the optimization procedure necessitates $O(n2^c + cn^2)$ operations where n denotes the dataset size and c is the number of clusters. The update process of mass membership $\mathbf{M}$ of our proposition (MECMdd-RWL) is similar to ECMdd. For a given $n \times n$ dissimilarity matrix, in this step ECMdd have $O(n2^c)$ complexity and as MECMdd-RWL introduces an additional sum on p terms, then its complexity at this step is $O(p \times n2^c)$. With the same approach the complexity for updating the prototypes and calculating the dissimilarity between objects and classes is $O(n2^c + cn^2)$ for ECMdd, thus $O(p \times (n2^c + cn^2))$ for MECMdd-RWL. The step of computation of the best relevance weight vectors needs the computation of 2^c denominators, the

Algorithm 1: MECMdd with local relevance weigh algorithms

Input: p dissimilarity matrices.

Parameters:

c number of clusters

α weighting exponent for cardinality

$\beta > 1$ weighting exponent

$\delta_l > 0$ dissimilarity between any object to the empty set

$\eta > 0$ coefficient for outlier identification

$\gamma \in [0, 1]$ coefficient to adjust the imprecise contribution

Initialization: $\mathbf{V}^0$ and $\mathbf{\Lambda}^0$

while not converge **do**

- (1) Compute the prototypes of meta-clusters utilizing (19) and (20);
- (2) Compute the credal partition matrix $\mathbf{M}$ using (8) and (9);
- (3) Compute the best relevance weight vector $\mathbf{\Lambda}$ according to (14);
- (4) Compute the new prototype set $\mathbf{V}$ using (18);

end

Output: The optimal $\mathbf{M}$, $\mathbf{V}$ and $\mathbf{\Lambda}$

computation of the numerator just once, and to repeat that for the p vector of relevance weights. Therefore, the total time complexity of this step is $O(n \times p \times 2^c + p \times 2^c) = O(n \times p \times 2^c)$. In summary, globally these steps cost $O(p \times n 2^c)$. Therefore, if the clustering process needs t iterations to converge, the total time complexity of MECMdd-RWL algorithm is $O(t \times p \times n 2^c)$.

B. MECMdd with relevance weight of each dissimilarity matrix estimated globally

The MECMdd-RWL algorithm can suffer numerical instability in the computation of the relevance weight of each dissimilarity matrix in each credal cluster when the algorithm produces single clusters or clusters containing objects with zero dissimilarity between them, i.e. $\sum_{i=1}^n m_{ij}^\beta \tau_{ijl} \rightarrow 0$. Indeed, this term is at the denominator of the weight update terms, Eq. (14), and if it becomes too small, the value of the fraction tends to infinity. We propose the variant MECMdd with global relevance weight (MECMdd-RWG) to provide a credal partition and prototype for each cluster, while learning relevance weights for all clusters in each view.

1) *Formulation:* Given the same formulation to Section IV-A1. MECMdd-RWG aims to find a partition $\mathbf{M}$, the corresponding prototypes $\mathbf{V}$, and a single relevance weight vector $\mathbf{\lambda} \in \mathbb{R}^p$. The objective function of MECMdd-RWG can be represented as follows:

$$J_{MECMdd-RWG}(\mathbf{M}, \mathbf{V}, \mathbf{\lambda}) = \sum_{i=1}^n \sum_{A_j \neq \emptyset} |A_j|^\alpha m_{ij}^\beta \sum_{l=1}^p (\lambda_l)^s \tau_{ijl} + \sum_{l=1}^p \delta_l^2 \sum_{i=1}^n m_{i\emptyset}^\beta. \quad (21)$$

Where $\sum_{l=1}^p (\lambda_l)^s \tau_{ijl}$ represents the global correspondence between an object $e_i \in E$ and the cluster prototype $v_k^\Omega \in E$, parameterized by $s > 1$ and by the weight vector $\mathbf{\lambda} = (\lambda_1, \dots, \lambda_p)$. The objective function $J_{MECMdd-RWG}$ is optimized using the same approach as in Section IV-A and under the constraints of Eq. (5), (6) and (22).

$$\lambda_l \in [0, 1], \quad \sum_{l=1}^p \lambda_l = 1, \quad \forall A_j \subseteq \Omega. \quad (22)$$

2) *Optimization:* We propose an efficient iterative approach to minimize (21). We utilize the Lagrange optimization technique to update one variable alternately.

Updating $\mathbf{M}$ while fixing $\mathbf{V}$ and $\mathbf{\lambda}$: During this step, both the prototype vector $\mathbf{V}$ and the relevance weight vector $\mathbf{\lambda}$ remain fixed.

Theorem IV.4. *The calculation of the credal partition, represented as $\mathbf{M}$, which minimizes the clustering objective function $J_{MECMdd-RWG}$ w.r.t. constraints (5) and (6) is derived from the following expression:*

$$m_{ij} = \frac{(|A_j|^\alpha \sum_{l=1}^p (\lambda_l)^s \tau_{ijl})^{-\frac{1}{\beta-1}}}{\sum_{A_k \neq \emptyset} (|A_k|^\alpha \sum_{l=1}^p (\lambda_l)^s \tau_{ikl})^{-\frac{1}{\beta-1}} + (\sum_{l=1}^p \delta_l^2)^{-\frac{1}{\beta-1}}}. \quad (23)$$

$$m_{i\emptyset} = 1 - \sum_{A_j \neq \emptyset} m_{ij}. \quad (24)$$

Proof. The proof proceeds in the same manner as that presented in Theorem IV.1. We utilize the Lagrangian multipliers, denoted as ξ , to address the constrained minimization problem concerning $\mathbf{M}$. Lagrangian can be written as follows:

$$\mathcal{L}(\mathbf{M}, \xi) = J_{MECMdd-RWG}(\mathbf{M}, \mathbf{V}, \mathbf{\lambda}) - \sum_{i=1}^n \xi_i \left(\sum_{\substack{A_j \subseteq \Omega, \\ A_j \neq \emptyset}} m_{ij} + m_{i\emptyset} - 1 \right). \quad (25)$$

By applying differentiation to the Lagrangian with respect to the variables m_{ij} , $m_{i\emptyset}$, and ξ_i , and subsequently setting the resulting derivatives to zero, we obtain the optimal solutions for m_{ij} and $m_{i\emptyset}$.

Updating $\mathbf{\lambda}$, while fixing $\mathbf{M}$ and $\mathbf{V}$:

Theorem IV.5. *Considering the relevance weight matching function w.r.t. constraints 22 and the weight vectors $\mathbf{\lambda} = (\lambda_1, \dots, \lambda_p)$ is determined using (21):*

$$\lambda_l = \frac{\left(\sum_{i=1}^n \sum_{A_j \neq \emptyset} |A_j|^\alpha m_{ij}^\beta \tau_{ijl} \right)^{-\frac{1}{s-1}}}{\sum_{h=1}^p \left(\sum_{i=1}^n \sum_{A_j \neq \emptyset} |A_j|^\alpha m_{ij}^\beta \tau_{ijh} \right)^{-\frac{1}{s-1}}}. \quad (26)$$

Proof. To demonstrate Theorem IV.5, we utilize the Lagrange multiplier, represented as ξ , to address the constrained minimization issue concerning λ , (21) can be defined as follows:

$$\mathcal{L}(\lambda, \xi) = J_{MECMdd-RWG}(\mathbf{M}, \mathbf{V}, \lambda) - \sum_{A_j} \xi_j \left(\sum_{l=1}^p \lambda_l - 1 \right). \quad (27)$$

The next of the proof proceeds in the same way as the Theorem IV.2. By taking the derivatives of the Lagrangian with respect to λ and ξ variables and setting the derivatives to zero, we obtain (26).

Updating $\mathbf{V}$ while fixing $\mathbf{M}$ and λ :

Theorem IV.6. *With the credal partition matrix $\mathbf{M}$ and dissimilarity matrix weight vector λ fixed, the prototype of the singleton cluster $e_p = v_k^\Omega$ of $A_j = \{\omega_k\}$ with $e_p \in E$, which minimizes the clustering objective function $J\text{-MECMdd-RWG}$, is first computed according to (28). Following this, the prototype of the imprecise cluster e_l , denoted as $v_k^{2\Omega}$, associated with $A_j \subseteq \{\omega_1, \dots, \omega_k\}$, where $e_l \in E$, is determined based on equations (19) and (20).*

$$p = \underset{1 \leq h \leq n}{\operatorname{argmin}} \sum_{i=1}^n m_{ij}^\beta \sum_{l=1}^p (\lambda_l)^s \tau_l(e_i, e_h). \quad (28)$$

Proof. The proof of Theorem IV.6 is straightforward according to [12].

3) *Computational algorithm and complexity:* For ease of implementation, we provide a summary of the proposed method in Algorithm 2. Following a similar reasoning as in Section IV-A3, we can conclude that the total time complexity of MECMdd-RWG is also of the order of $O(t \times p \times n2^c)$ and the optimization is converging.

Algorithm 2: MECMdd with global relevance weigh algorithms

Input: p dissimilarity matrices.

Parameters:

Refer to parameters of Algorithm 1

Initialization: $\mathbf{V}^0$ and λ^0

while not converge **do**

- (1) Compute the prototypes of meta-clusters utilizing (19) and (20);
- (2) Compute the credal partition matrix $\mathbf{M}$ using (23) and (24);
- (3) Compute the best relevance weight vector λ according to (26);
- (4) Compute the new prototype set $\mathbf{V}$ using (28);

end

Output: The optimal $\mathbf{M}$, $\mathbf{V}$ and λ

Remark. *It is interesting to note that, numerical instability, such as overflow or division by zero, can always arise when calculating the significance weight of each dissimilarity matrix. This typically happens when the algorithm generates credal clusters such that $\sum_{i=1}^n \sum_{A_j \neq \emptyset} m_{ij}^\beta d_{ijl} \rightarrow 0$. However, the likelihood of these types of numerical inconsistencies occurring is greater for the algorithms discussed in Section*

IV-A. Additionally, the exponential growth in the number of clusters significantly increases computational complexity, making it challenging to interpret the assignment of objects to high-cardinality meta-clusters. To address this, constraints can be imposed to limit the number of focal elements in the meta-cluster. For instance, similar to ECM [7], we may restrict the meta-cluster to contain only two clusters.

V. EMPIRICAL RESULTS

In this section, experiments on twelve real-world datasets were conducted to evaluate the performance of proposed algorithms. First, the detailed description of datasets, and comparison process are introduced in section V-A. Then, in section V-B, we investigate the performances of the proposed algorithms, comparing them against each other by an application to medical image segmentation using different distances functions. The performance of our approaches against related and state-of-the-art methods is evaluated in section V-C. Finally, section V-D analyzes the parameters sensitivity of our methods.

Remark. *All experimental data and code are available in a public GitHub repository¹.*

A. Experimental setting

To compare the performance of partitions generated by the proposed multi-view relational clustering algorithms (MECMdd-RWL and MECMdd-RWG) against each other, and to evaluate the performance of the proposed clustering algorithm in comparison with other methods, we performed experiments on twelve publicly available real datasets. Data includes simple vector data, multi-view data, categorical data and multivariate time series data. The state-of-the-art methods used to compare the performance of our methods consist of two classical single-view relational clustering methods, namely FCMdd [24] and ECMdd [12], a vector single-view ECM [7], and three representative multi-view relational and vector clustering methods, which include MRDCA [19], MRFCMdd [22], and WMVEC [17].

For simple vector data, we considered four benchmark datasets selected from the UCI machine learning repository [33], including Thyroid gland, Ecoli, Abalone, and Wine. The different datasets are described in Table II, where n represents the number of objects, p represents the number of variables, and c represents the number of predefined clusters.

TABLE II: Summary of the simple vector data.

Datasets	n	p	c
Thyroid gland	215	5	3
Ecoli	336	7	8
Abalone	4177	8	3
Wine	178	13	3

¹The repository can be accessed at <https://github.com/armelsoubeiga/MECMdd>.

For single-view relational algorithms, a dissimilarity matrix was computed by considering all p attributes simultaneously using the standard Euclidean distance. For multi-view relational algorithms, multiple dissimilarity matrices were used, each computed by considering only one attribute at a time using the standard Euclidean distance. Multi-view vector algorithms and single-view vector algorithms were applied to the raw data, treating each attribute as a separate view or combining all attributes into one table, respectively.

The basic information of the used multi-view datasets are shown in Table III. We used two publicly available multi-view datasets, including Prokaryotic Phyla [34] and Multiple features [35]. The Prokaryotic Phyla dataset includes 551 prokaryotic species, characterized by three multi-view data such as text and two genomics representations. Multiple features dataset is available from the UCI repository². This dataset contains 2000 handwritten digits from Dutch utility maps, each digit represented by six feature view sets, we utilized three sets: Fourier coefficients, profile correlations, and Karhunen-Love coefficients as in [35]. For each view in a multi-view dataset, one dissimilarity matrix was computed considering all attributes in the view, according to the Euclidean distance. Single-view clustering algorithms were applied to the Euclidean matrix obtained by considering all attributes across all views.

TABLE III: The details of the used multi-view datasets

Datasets	n	c	p	Views name	Features
Multiple features	2000	10	3	Fourier coefficients	76
				Profile correlations	216
				Karhunen-Love coefficients	64
Prokaryotic Phyla	551	4	3	Gabor	393
				Wavelet moments	3
				CENTRIST	438

In order to highlight the potential for broader application and to validate the proposed multi-view relational clustering approaches, we also utilized categorical and time series real-world datasets. In Table IV, we show the characteristics of three categorical datasets of different size available on the UCI Machine Learning repository : Car Evaluation, Breast Cancer, and Zoo. In experiments, we transform the categorical datasets to numerical with the one-hot encoding technique in order to use the numerical vector methods ECM and WMVEC [36]. For ECM, all new derived attributes are used in the algorithm within a single view. For mean-based multi-view method, WMVEC, the new attributes from each transformed variable are considered as a separate view. To adapt categorical data into a relational context, categorical data dissimilarity measures can be used. In the literature, several categorical measures have been proposed (see [37] for a review). In this experiment, the most used dissimilarity measures are considered : Hamming distance [38] for binary categorical

variables, and Gower distance [39] for categorical variables with more than two modes. Based on these dissimilarity measures, medoid-based single-view clustering methods (FCMdd and ECMdd) are applied to the dissimilarity matrix obtained by considering all attributes. For medoid-based multi-view clustering algorithms (MRDC, MRFCMdd and MECMdd), dissimilarity matrices are calculated by considering each categorical variable as a separate view.

TABLE IV: Datasets information of the used categorical datasets.

Datasets	n	p	c
Car Evaluation	1728	6	4
Breast Cancer	286	9	2
Zoo	101	16	7

In Table V, we show the characteristics of the three multivariate time series datasets selected. The multivariate continuous time series datasets are available online at the UCR time series [40]. The biofam dataset is multivariate categorical time series from the data of the retrospective biographical survey carried out by the Swiss Household Panel [41]. As shown in this Table, datasets with different number of time series, different length of time series, and different number of classes are selected. In this experiments, we use Dynamic Time Warping (DTW) [42] distance for clustering continuous time series data. DTW is the most well known technique for evaluating similarity/dissimilarity of time series with respect to their shape information. Unlike the Euclidean distance, DTW can compare time series of variable size and is robust to shifts or dilatation across the time dimension. The dissimilarity between pairs of categorical time series is calculated using the Optimal Matching (OM) [43] measure. Each univariate time series is considered as a view. The single-view medoid-based clustering algorithms were applied to the dissimilarity matrix obtained by a linear sum combination, considering all dissimilarities of all views.

TABLE V: Multivariate time series selected for experimental studies.

Datasets	n	p	c	Length	Type of features
UWaveGestureLibrary	120	3	8	315	Continuous
StandWalkJump	12	4	3	2500	Continuous
Biofam dataset	2000	3	2	30	Categorical

All dissimilarity matrices were normalized according to their overall dispersion [44] to have the same dynamic range. This means that each dissimilarity $d(e_k, e_{k'})$ in a given dissimilarity matrix has been normalized as $d(e_k, e_{k'})/T$, where $T = \sum_{k=1}^n d(e_k, g)$ is the overall dispersion and $g = e_l \in E = \{e_1, \dots, e_n\}$ is the overall prototype, which is computed according to $l = \operatorname{argmin}_{1 \leq h \leq n} \sum_{k=1}^n d(e_k, e_h)$.

The relational and vector hard clustering algorithms used in the comparison were respectively applied to the similarity matrices or vector data to obtain a set of groups $\mathcal{G} = (G_1, \dots, G_c)$. From all fuzzy partitions, we generate hard partitions with the maximum principle rule i.e.

²<https://archive.ics.uci.edu/dataset/72/multiple+features>

by defining the hard cluster $G_k (k = 1, \dots, c)$ as $G_k = \{\mathbf{x}_i : u_{ik} \geq u_{im} \ \forall m \in \{1, \dots, c\}\}$. Where the quantity u_{ik} represents the membership degree of object $x_i (i = 1, \dots, n)$ in fuzzy cluster $k (k = 1, \dots, c)$. For the proposed algorithms and credal partitions used in the comparison, we can employ pignistic transformation [30], [45] BetP in Eq. (1) to turn the credal partition into a fuzzy partition. With such conversion, the mass of belief of the meta-cluster will be evenly distributed to the relevant singleton clusters. Each clustering algorithm was run (until the convergence to a stationary value of the adequacy criterion) 2 times and the best result was selected according to the adequacy criterion and the parameters presented in Table VII.

B. Comparison of proposed algorithms

In this section, the strength of MECMdd is illustrated using an example in medical imaging taken from [46]. We compare the two variants: when the weights are estimated for each cluster locally (MECMdd-RWL) versus when they are estimated globally (MECMdd-RWG). An image of a pathological brain of tumor was acquired using magnetic resonance Imaging. Three main areas can be distinguished: the brightest area corresponds to a pathological region, the gray area represents normal brain tissues, and the other small white areas correspond to cerebrospinal fluid. The goal was to isolate the pathological area from the other parts of the brain by partitioning into $c = 2$ clusters.

To facilitate the computation, the image was transformed into Grayscale and reduced to 128×128 pixels. We then derived eight distance and similarity measures from the pixel coordinates and their values of the image. The similarity and distance functions considered are: Mahalanobis, cosine, correlation, Dice, Euclidean, Wave, Minkowski and Hellinger [47]. Indeed, according to [47], not all similarity measures for segmentation have the same properties, depending on segment boundaries, region-sensitive refinement and compensation for differences in segment ambiguity. Thus, collaborating with multiple distances while determining relevance weights for each is a considerable asset to the task. Clustering was performed simultaneously on these distance matrices and the parameters defined in Table VII for the MECMdd algorithm were iterated. At each iteration, the credal partition is transformed into a hard partition by the pignistic approach and the silhouette index adapted to multi-view relational data represented by p dissimilarity matrices is calculated [15]. Figure 2 summarizes the experimental pipeline and the results obtained. The performances of the best silhouettes are: $SI = 0.71$, $\beta = 1.8$ for MECMdd-RWL and $SI = 0.67$, $\beta = 2$ for MECMdd-RWG, with both algorithms using $\alpha = 1$, $\gamma = 0$, and $\eta = 1$.

When we observe Figure 2 (a) versus Figure 2 (b), MECMdd-RWL, with an $SI = 0.67$, identified ambiguous zones correctly, demonstrating effective management of subtle variations. On the other hand, MECMdd-RWG, with an $SI = 0.71$, classified certain zones incorrectly, indicating less

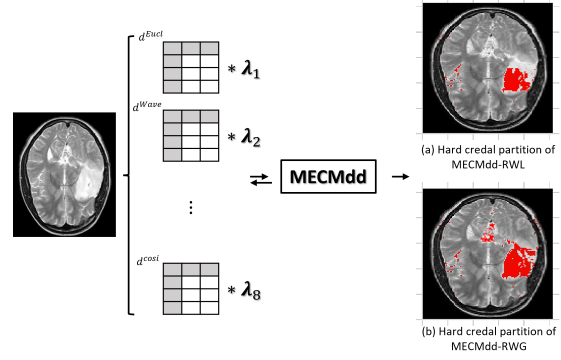


Fig. 2: Hard credal partition obtained from MECMdd-RWL and MECMdd-RWG with an adaptive weight. Red corresponds to the pathology area.

efficiency in dealing with specific ambiguities despite superior overall performance. Table VI shows that the MECMdd-RWL algorithm considers a single matrix (d^{Dice}) when clustering normal brain tissue areas, while four matrices are considered when clustering pathological areas. However, all our proposed algorithms correctly identify areas of uncertainty and distinguish intermediate areas, while assigning a weight to each distance.

TABLE VI: Vectors of relevance weight of brain image segmentation using MECMdd.

Data matrix	MECMdd-RWL		MECMdd-RWG
	Tissues area	Pathological area	
$d^{Mahalanobis}$	0.00	0.00	0.00
$d^{Cosinus}$	0.00	0.26	0.53
$d^{Correlation}$	0.00	0.13	0.29
d^{Dice}	1.0	0.50	0.11
$d^{Euclidean}$	0.00	0.00	0.00
d^{Wave}	0.00	0.00	0.01
$d^{Minkowski}$	0.00	0.00	0.01
$d^{Hellinger}$	0.00	0.11	0.06

C. Comparison with other algorithms

To assess the performance of the different clustering algorithms and compare them, we assume that the ground truth is known, and the hard partition obtained from each algorithm on each dataset was compared with the predefined partition. The parameters of different methods are presented in Table VII, and the comparison criteria used were the Adjusted Rand Index (ARI) [48] and the Precision (P). Higher values of these two indices indicate better clustering results. The higher value of each index is kept according to the number of iterations (2 times) and hyper-parameters of each algorithm (Table VII). Based on the analysis of evidential clustering parameters in [7], [12], [18], α have little impact on the results. In the rest of the work, we have renamed the following datasets UWaveGestureLibrary (UGL), StandWalkJump (SWJ) and Multiple features (mfeat).

TABLE VII: The parameters setting of different algorithms. $Q_{0.95}$ is the quantile function at 95%, p number of views.

Algorithms	Parameters
ECM	$\alpha = 1; \beta \in \{1.1, 1.2, \dots, 2.0\}; \delta = 10$
ECMdd	$\alpha = 1; \beta \in \{1.1, 1.2, \dots, 2.0\}; \delta \in \{10, Q_{0.95}(\tau_{ij:i \neq j})\}; \eta \in \{1, 2\}; \gamma \in \{0, 1\}$
MECMdd	$\alpha = 1; \beta \in \{1.1, 1.2, \dots, 2.0\}; \delta^p \in \{10, Q_{0.95}(\tau_{ij:i \neq j})\}; \eta \in \{1, 2\}; \gamma \in \{0, 1\}; s = 2$
WMVEC	$\alpha = 1; \beta \in \{1.1, 1.2, \dots, 2.0\}; \lambda \in \{e^0, e^5, e^{10}\}$
FCMdd	$m \in \{1.1, 1.5, 2, 2.5, 3.0\}$
MRFCMdd	$s = 2$
MRDC	<i>none</i>

Table VIII present the performance of the clustering algorithms on each dataset. Performance was evaluated using the ARI and Precision (P) criteria. Compared with single-view clustering methods (i.e. ECM, ECMdd and FCMdd), MECMdd, MRFCMdd, WMVEC generally has obvious advantages on the multi-view datasets (mfeat and prokaryoti), which indicates that it is not appropriate to use single-view clustering methods to deal with multi-view data. We can also observe that, in general, relational algorithms perform better than vector algorithms, the reason being that relational methods use an adapted distance measure, unlike the Euclidean distance used with vector algorithms. In particular, while medoid-based relational clustering approaches can cluster row data time series using adapted measures, vector-based algorithms are not applicable (Table VIII). In general, variants of algorithms that locally estimate the weights of the views yield better results. Significantly, MECMdd-RWL stands out not just for its capability to transition into hard or fuzzy partitions, resulting in remarkable performance, but also for its ability to provide a deeper understanding of the data when utilized with a credal partition.

D. Analysis of parameters

Before running MECMdd, parameters δ , β , α , η , γ and s should be set. Parameters δ , β and α have the same meanings as those in ECM, and η and γ are similar to ECMdd. In ECMdd, the parameter γ determines how much uncertainty influences the dissimilarity between objects and imprecise clusters. The parameter η helps differentiate outliers from potential medoids when identifying the prototypes of meta-classes. As suggested in [12], a value can be used as a starting default one but it can be modified according to what is expected from the user. The choice δ is more difficult and is strongly data dependent, we suggest using the default distance threshold for each view (similarity matrix), where only 5% of distances are located.

VI. CONCLUSION

In this paper, we propose medoid-based relational multi-view evidential clustering algorithms that are able to partition objects taking into account simultaneously their relational

descriptions given by multiple dissimilarity matrices. Referred to as MECMdd, it is based on the framework of belief functions. MECMdd is able to address the uncertainty and imprecision of multi-view relational data clustering, and provides a credible partition, which extends fuzzy partition. It can automatically learn the importance of each view by estimating weights locally for each cluster (MECMdd-RWL) and globally (MECMdd-RWG) in a collaborative learning framework. Experimental results obtained on several real datasets demonstrate the potential and superiority of our proposed algorithm. In future work, we plan to extend the proposed method to handle other constraints on weights and also the possibility to identify several medoids for each cluster with different weights.

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TABLE VIII: Clustering performance of hard, fuzzy, and credal partitions algorithms with different relational datasets measured on ARI and Precision (P) indexes.

Data	ecm		ecmdd		mecmdd-rl		mecmdd-rwg		wmvec		fcmdd		mrfcmdd-rl		mrfcmdd-rwg		mrdca-rl		mrdca-rwg	
	ARI	P	ARI	P	ARI	P	ARI	P	ARI	P	ARI	P	ARI	P	ARI	P	ARI	P	ARI	P
abalone	0.59	0.41	0.62	0.44	0.62	0.44	0.63	0.44	0.66	0.45	0.61	0.42	0.62	0.44	0.62	0.44	0.62	0.43	0.62	0.43
ecoli	0.80	0.77	0.72	0.64	0.82	0.77	0.72	0.73	0.70	0.68	0.78	0.65	0.36	0.29	0.79	0.71	0.80	0.73	0.76	0.60
wine	0.72	0.59	0.73	0.60	0.76	0.63	0.75	0.63	0.72	0.59	0.72	0.59	0.64	0.47	0.66	0.50	0.85	0.78	0.71	0.55
thyroid	0.53	0.59	0.61	0.68	0.95	0.93	0.64	0.70	0.63	0.69	0.55	0.69	0.57	0.65	0.55	0.61	0.52	0.57	0.60	0.66
prokaryoti	0.48	0.45	0.50	0.44	0.65	0.50	0.59	0.48	0.62	0.53	0.58	0.46	0.59	0.47	0.60	0.49	0.62	0.53	0.60	0.48
mfeat	0.90	0.47	0.92	0.41	0.92	0.35	0.93	0.57	0.92	0.58	0.61	0.30	0.87	0.39	0.86	0.33	0.92	0.64	0.92	0.63
car	0.51	0.61	0.53	0.63	0.56	0.66	0.51	0.71	0.56	0.70	0.52	0.62	0.54	0.62	0.49	0.56	0.53	0.66	0.70	0.67
cancer	0.55	0.63	0.51	0.58	0.57	0.64	0.57	0.58	0.54	0.58	0.50	0.58	0.58	0.40	0.55	0.56	0.61	0.55	0.59	0.63
zoo	0.90	0.92	0.85	0.76	0.93	0.92	0.91	0.92	0.67	0.33	0.91	0.89	0.82	0.77	0.63	0.80	0.90	0.88	0.89	0.75
biofam	-	-	0.41	0.58	0.44	0.58	0.43	0.44	-	-	0.51	0.52	0.45	0.55	0.42	0.33	0.50	0.42	0.50	0.51
UGL	-	-	0.86	0.44	0.89	0.52	0.87	0.46	-	-	0.87	0.46	0.88	50	0.87	0.47	0.89	0.51	0.84	0.39
SWJ	-	-	0.65	0.37	0.65	0.37	0.59	0.32	-	-	0.56	0.21	0.55	0.29	0.59	0.27	0.55	0.23	0.50	0.22

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